

Early Rotorcraft

War set back helicopter development in Europe; Germany built several dozen helicopters of different types in the 1940s but production was limited by lack of resources. In North America, pioneers like Igor Sikorsky, Frank Piasecki, Arthur Young, and Stanley Hiller drove progress, producing the forerunners of the machines flown today. The first-ever helicopter winch rescue took place in 1945 when a Sikorsky R-5 flown by Igor's son-in-law lifted two men off a sinking barge in a storm.

▷ Sikorsky R-4 1942

Origin USA

Engine 1 x 200 hp Warner R-500-3 Super Scarab radial

Top speed 75 mph (120 km/h)

This was the breakthrough machine. Sikorsky's R4, developed from the VS-300, was the first truly practical helicopter and was mass-produced for the American and British armed forces toward the end of WWII.



△ Kellett XO-60 autogyro 1943

Origin USA

Engine 1 x 330 hp Jacobs R-915-3 radial

Top speed 125 mph (201 km/h)

The last gasp of the autogyro before the helicopter eased it out, the XO-60 had a rotor that could be driven by the engine for jump takeoffs but was demanding to fly and suffered several accidents.



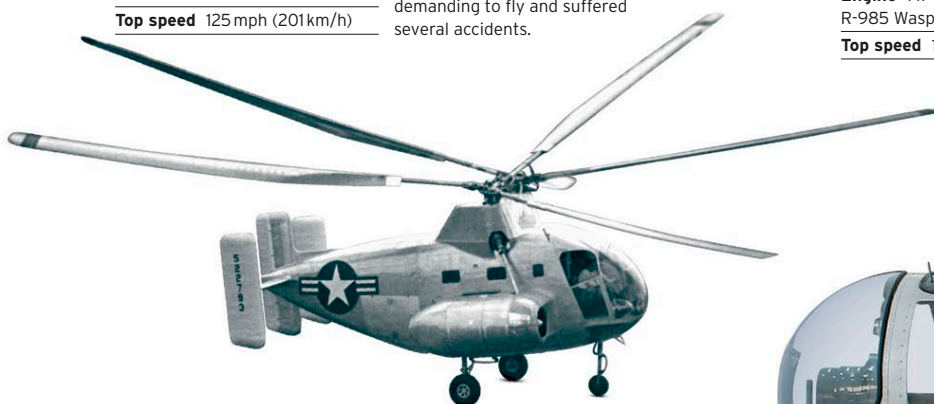
△ Sikorsky S51/H-5 1945

Origin USA

Engine 1 x 450 hp Pratt & Whitney R-985 Wasp Junior radial

Top speed 106 mph (171 km/h)

This was the second-generation Sikorsky, designed with postwar civilian uses in mind. It was too complex and expensive for the nonmilitary market; the US Navy remained the largest customer.



△ Kellett XR-10 1947

Origin USA

Engine 2 x 415 hp Continental R-975-15 radial

Top speed 100 mph (161 km/h)

Kellett's 5-ton twin-engined transport helicopter was ambitious, but mechanical failure of the intermeshing rotors killed the company's test pilot and the project was abandoned.



△ P-V Engineering Forum PV-2 1943

Origin USA

Engine 1 x 90 hp Franklin air-cooled opposed

Top speed 100 mph (161 km/h)

P-V eventually morphed into Boeing Vertol, makers of the mighty Chinook; Frank Piasecki's experimental PV-2 had full cyclic and collective rotor pitch control and tail rotor antitorque.



◁ Focke Achgelis Fa-330 1943

Origin Germany

Engine None

Top speed 25 mph (44 km/h)

Heinrich Focke was sacked from Focke-Wulf for being politically suspect, but the Nazis allowed him to work on helicopters. The Fa-330 was an unpowered submarine-towed rotary kite used for ship-spotting.

